

To implement elbow method to find best k

k range = range(1, 10)
sse = []

for k in k range:

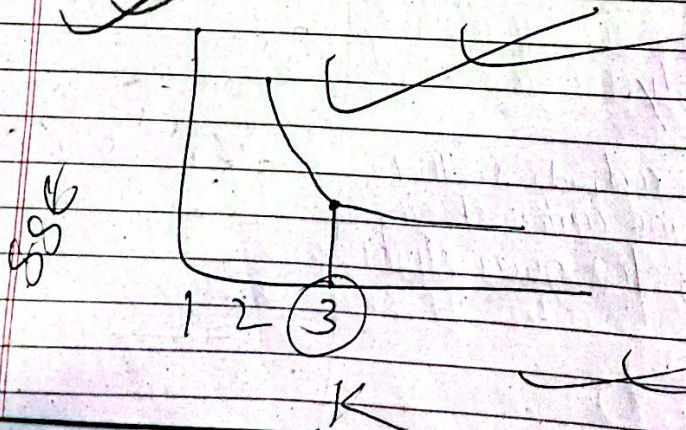
random state = 42

km = KMeans(n_clusters=k,
km.fit(df[["Age", "Income (\$)"]])
sse.append(km.inertia_)

provides sum of squared error

Remember to use random state only once for all runs

So, plt.xlabel("k")
plt.ylabel("sum of squared error")
plt.plot(k range, sse)



K-Means Clustering (Numerical)

Suppose that the data mining task is to cluster points into three clusters, (k=3)

where the points are

A1(2,10), A2(2,5), A3(8,4), B1(5,8),
B2(7,5), B3(6,4), C1(1,2), C2(4,9)

Distance function = Euclidean

And A1, B1, & C1 as the center of each cluster respectively.

Iter/F

$$\text{distance}(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Iteration - 1

Point x_1, y_1	(2,10) Centroid 1	(5,8) Centroid 2	(1,2) Centroid 3	Cluster
A1 (2,10)	0	3.606	8.062	cluster 1
A2 (2,5)	5	4.243	3.162	cluster 3
A3 (8,4)	8.485	5	7.280	cluster 2
B1 (5,8)	3.606	0	7.211	cluster 2
B2 (7,5)	7.071	3.606	6.708	cluster 2
B3 (6,4)	7.211	4.123	5.385	cluster 2
C1 (1,2)	8.06	7.21	0.00	cluster 3
C2 (4,9)	2.24	1.41	7.62	cluster 2

Now, we need to recompute the new cluster centroids of each cluster (means)

for cluster 1:- $(2, 10)$ directly

for cluster 2:-

$$\text{Centroid 2} = \left(\frac{8+5+7+6+4}{5}, \frac{4+8+5+4+9}{5} \right)$$

$$= (6, 6)$$

for cluster 3:-

$$\text{Centroid 3} = \left(\frac{2+1}{2}, \frac{5+2}{2} \right)$$

$$= (1.5, 3.5)$$

Iteration 2

Compare all the some clusters

K-means ++

Cluster the following data points into 3 clusters, $(7, 4), (8, 3), (5, 9), (3, 3), (4, 3), (10, 1)$ using K-Means ++ Algorithm.

let $C_1 = (7, 4)$ (choose one randomly)

Step 1:- Finding distance from C_1 to x_i .

$$d(C_1, P_1) = 0$$

$$d(C_1, P_2) = \sqrt{2} = 1.414$$

$$d(C_1, P_3) = \sqrt{29} = 5.385$$

$$d(C_1, P_4) = \sqrt{17} = 4.123$$

$$d(C_1, P_5) = \sqrt{37} = 6.083$$

$$d(C_1, P_6) = 3\sqrt{2} = 4.243$$

X	$\text{Mind}(C_i, P_j)^2$
P_1	0
P_2	2
P_3	29
P_4	17
P_5	37
P_6	18
Total	103

Square

Step 2: - finding probabilities as

X	Prob
P_1	—
P_2	$2/103$
P_3	$29/103$
P_4	$17/103$
P_5	$37/103$ highest prob
P_6	$18/103$

The probability of choosing next centroid point C_2 is $P_5 (1,3)$.

$$\text{So, } C_2 = (1,3)$$

Now,
finding distance from C_1 & C_2 to x .

For P_1

$$d(C_1, P_1) = 0$$

$$d(C_2, P_1) = \sqrt{37}$$

For P_2

$$d(C_1, P_2) = \sqrt{2}$$

$$d(C_2, P_2) = 7$$

for P_3 ,

$$d(C_1, P_3) = \sqrt{29}$$

$$d(C_2, P_3) = 2\sqrt{13}$$

for P_4 ,

$$d(C_1, P_4) = \sqrt{17}$$

$$d(C_2, P_4) = 2$$

for P_5 ,

$$d(C_1, P_5) = \sqrt{37}$$

$$d(C_2, P_5) = 0$$

for P_6

$$d(C_1, P_6) = 8\sqrt{2}$$

$$d(C_2, P_6) = \sqrt{85}$$

X	$\text{Min}(d(x))^2$
P_1	— 0
P_2	$\min(2, 49) = 2$
P_3	$\min(29, 52) = 29$
P_4	$\min(17, 4) = 4$
P_5	$\min(37, 0) = 0$
P_6	$\min(18, 85) = 18$
Total = 53	

Now,
 finding probabilities of

X	Prob
P_1	—
P_2	$2/53$
P_3	$29/53$ Highest
P_4	$4/53$
P_5	—
P_6	$18/53$

So, Next centroid is P_3

$$\begin{aligned} \text{ie.} \\ C_3 &= P_3 \\ &= (5, 9) \end{aligned}$$

Initial

$$\begin{aligned} \text{So, Centroids} &= C_1 = (7, 4) \\ &= C_2 = (4, 3) \\ &= C_3 = (5, 9) \end{aligned}$$